UNITED STATES OF AMERICA
DEPARTMENT OF EDUCATION
UNIVERSITY OF LaTeX

ΠΑΝΠΙΣΤΗΜΙΟ LaTeX
LaTeX UNIVERSITY

SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

**Master's Thesis**
**Field : Computer Science**
**Specialization : Artificial Intelligence**

---

# Geodesics Shadows and Ringdown Signatures of Schwarzschild and Kerr Black Holes

---

**Submitted by:**
ZEDX Pro

**Supervised by:**
Prof. FRANK Mittelbach

**Thesis Committee:**
Prof. LESLIE Lamport
Prof. DONALD Knuth

Academic Year 2025-2026

# DEDICATION

*To the entire TeX and LaTeX community, whose knowledge, dedication, and generosity have made scientific writing more accessible to everyone.*

**— ZEDX Pro**

# ACKNOWLEDGMENTS

I sincerely thank **God** for His guidance, strength, and countless blessings.

I also express my sincere appreciation to **Prof. Donald E. Knuth**, **Prof. Leslie Lamport**, and **Prof. Frank Mittelbach** for their outstanding contributions to the development of TeX and LaTeX, which have made scientific writing more accessible and elegant.

Abstract

This work presents a theoretical and numerical study of exact solutions of the Einstein field equations, with particular emphasis on the Schwarzschild and Kerr metrics. After recalling the geometric foundations of general relativity, the geodesic structure of these spacetimes is analyzed, including the innermost stable circular orbit and its dependence on the black hole spin. A numerical ray-tracing procedure is then developed to integrate null geodesics and compute gravitational lensing effects and the black hole shadow, whose predicted size is compared with Event Horizon Telescope observations of M87*. Finally, the dynamical response of these spacetimes to small perturbations is investigated through the quasinormal mode spectrum, providing a black hole spectroscopy framework that is illustrated using the ringdown of the gravitational-wave event GW150914. The consistency between the analytical, numerical, and observational results obtained throughout this work supports the Kerr metric as the standard description of astrophysical black holes.

**Keywords:** General relativity, Schwarzschild metric, Kerr metric, black hole, geodesics, gravitational lensing, quasinormal modes.

# Table of Contents

# List of Tables

# List of Figures

# Nomenclature

## Abbreviations

| Symbol | Meaning | Field |
|---|---|---|
| AI | Artificial Intelligence | General |
| CNN | Convolutional Neural Network | Deep Learning |
| ReLU | Rectified Linear Unit | Activation |
| SGD | Stochastic Gradient Descent | Optimization |
| GPU | Graphics Processing Unit | Hardware |

## Greek Letters

| Symbol | Meaning | Unit |
|---|---|---|
| $\eta$ | Learning rate | -- |
| $\lambda$ | Regularization parameter | -- |
| $\sigma$ | Sigmoid function | -- |

# Thesis Overview

The **General Introduction** presents the research context, the motivations behind this work, and the main objectives pursued throughout the thesis.

In **Chapter 1**, we recall the geometric foundations of general relativity, introduce the Einstein field equations, and derive the Schwarzschild solution, which serves as the reference exact solution used throughout this work.

**Chapter 2** extends this framework to rotating compact objects through the Kerr metric, presenting the geodesic equations, the associated conserved quantities, and the spin dependence of the innermost stable circular orbit.

**Chapter 3** implements a numerical ray-tracing procedure to integrate null geodesics in the Schwarzschild and Kerr geometries, allowing the computation of gravitational lensing effects and the black hole shadow, which is compared with Event Horizon Telescope observations of M87*.

**Chapter 4** addresses the dynamical response of these spacetimes to small perturbations, introducing the quasinormal mode spectrum and the black hole spectroscopy approach, illustrated through the ringdown analysis of the gravitational-wave event GW150914.

The **General Conclusion** summarizes the main findings of this work and outlines promising directions for future research.

**Appendix A** validates the numerical methods used in Chapters 3.1 and 4.1, comparing the ray-tracing and quasinormal mode codes against known analytical and reference results.

**Appendix B** collects complementary analytical derivations, including the Christoffel symbols of the Schwarzschild metric and the circular-orbit conditions in the Kerr geometry underlying the results of Chapters 1.1 and 2.1.

# 0

# Introduction

## General Context

General relativity, formulated by Einstein in 1915, remains the cornerstone theory describing gravitation as the curvature of spacetime induced by the presence of mass and energy [3]. Since its formulation, it has been confirmed by numerous experimental and observational tests, ranging from the perihelion precession of Mercury to the direct detection of gravitational waves [4].

The study of exact solutions to the Einstein field equations, such as the Schwarzschild and Kerr metrics, continues to play a central role in understanding compact astrophysical objects such as black holes and neutron stars [5].

## Problem Statement

Despite a century of theoretical and observational progress, several open questions remain, including the behavior of spacetime near singularities, the compatibility of general relativity with quantum mechanics, and the interpretation of dark energy within the cosmological constant framework [6]. This work addresses some of these questions by proposing a detailed study of specific exact solutions of the Einstein field equations and their physical implications.

## Objectives

The main objective of this thesis is to analyze, interpret, and numerically explore selected exact solutions of general relativity. Secondary objectives include studying the geodesic structure associated with these solutions and comparing their predictions with astrophysical observations.

> The main objective of this work is to conduct a detailed theoretical and numerical study of exact solutions of the Einstein field equations, with particular emphasis on the Schwarzschild and Kerr metrics, in order to analyze their geodesic structure, horizon properties, and physical consequences for the behavior of matter and light in strong gravitational fields.

## Scientific Context

The rapid development of relativistic astrophysics, particularly following the direct detection of gravitational waves and the imaging of black hole shadows, has made the study of strong-field gravity a highly active research area. From compact binary mergers to the dynamics of matter near event horizons, general relativity provides the essential theoretical framework for interpreting these phenomena.

Exact solutions of the Einstein field equations have emerged as indispensable tools for modeling astrophysical compact objects, owing to their ability to capture the nonlinear geometric structure of spacetime in regions of strong gravity. Their confirmation through observations such as the Event Horizon Telescope imaging of M87* has established them as central benchmarks of modern relativistic astrophysics [7].

*"Imagination is more important than knowledge."*

— Albert Einstein

# 1 Theoretical Framework of General Relativity

## 1.1 Introduction

This chapter introduces the theoretical foundations required for the study developed in this thesis. We first recall the geometric formulation of general relativity through the Einstein field equations, then present the Schwarzschild solution as the reference exact solution used throughout this work [3, 6].

## 1.2 Geometric Foundations

General relativity describes gravitation not as a force, but as a manifestation of spacetime curvature. The geometry of spacetime is encoded in the metric tensor $g_{\mu\nu}$, which defines the invariant interval

$$ds^2 = g_{\mu\nu}\, dx^\mu dx^\nu, \tag{1.1}$$

Greek indices run from 0 to 3, with $x^0 = ct$.

using the Einstein summation convention over repeated indices [5].

The curvature of spacetime is related to the distribution of matter and energy through the Einstein field equations

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{1.2}$$

where $R_{\mu\nu}$ is the Ricci tensor, $R$ the scalar curvature, and $T_{\mu\nu}$ the stress-energy tensor [6]. As stated in Eq. (1.2), the left-hand side is purely geometric while the right-hand side encodes the physical matter content, illustrating the interplay expressed by Wheeler's summary: matter tells spacetime how to curve, and spacetime tells matter how to move[1].

## 1.3 The Schwarzschild Solution

For a static, spherically symmetric mass distribution in vacuum ($T_{\mu\nu} = 0$), Eq. (1.2) admits the exact Schwarzschild solution [8]

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2\, d\Omega^2, \tag{1.3}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$ is the metric on the unit sphere. The quantity

$$r_s = \frac{2GM}{c^2} \tag{1.4}$$

defines the Schwarzschild radius, corresponding to the event horizon of a non-rotating black hole [9]. Equation (1.3) reduces to the flat Minkowski metric as $r \to \infty$, recovering special relativity far from the mass [10].

[1] This heuristic formulation, attributed to J. A. Wheeler, is widely used in introductory treatments of general relativity [5].

**Figure 1.1:** Embedding diagram illustrating the curvature of spacetime around a spherical mass, as described by the Schwarzschild metric (Eq. (1.3)).

## 1.4 Comparison of Exact Solutions

Table 1.1 summarizes the main properties of the exact solutions considered as reference cases in relativistic astrophysics.

**Table 1.1:** Comparison of classical exact solutions of the Einstein field equations.

| Solution | Symmetry | Charge | Angular momentum |
|---|---|---|---|
| Schwarzschild [8] | Spherical | No | No |
| Reissner--Nordström [11] | Spherical | Yes | No |
| Kerr [12] | Axial | No | Yes |
| Kerr--Newman [13] | Axial | Yes | Yes |

As shown in Table 1.1, the Schwarzschild metric of Eq. (1.3) corresponds to the simplest case, from which the more general rotating and charged solutions can be derived [12].

## 1.5 Conclusion

This chapter recalled the geometric foundations of general relativity, culminating in the derivation of the Schwarzschild solution (Eq. (1.3)), which serves as the reference framework for the analysis developed in the remainder of this thesis [3, 6].

*``Heat, like a fluid, tends to pass from a body which is hotter to one which is colder.''*

— Joseph Fourier

# 2 Geodesic Structure of Rotating Spacetimes: The Kerr Metric

## 2.1 Introduction

Building on the geometric foundations and the Schwarzschild solution presented in Chapter 1.1, this chapter extends the discussion to rotating compact objects through the Kerr metric [12]. We first derive the geodesic equations governing the motion of test particles and photons in a general spacetime, then present the Kerr solution in Boyer--Lindquist coordinates [14], and finally discuss its geodesic structure, including the photon sphere and the innermost stable circular orbit (ISCO) [15].

## 2.2 Geodesic Equations in General Relativity

The trajectories of freely falling particles and light rays in a curved spacetime are described by the geodesic equation

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0, \tag{2.1}$$

Massive particles follow timelike geodesics ($ds^2 < 0$), photons follow null geodesics ($ds^2 = 0$).

where $\Gamma^\mu_{\alpha\beta}$ are the Christoffel symbols associated with the metric $g_{\mu\nu}$ introduced in Eq. (1.1), and $\tau$ is an affine parameter along the trajectory [5]. For massive particles $\tau$ is taken to be the proper time, while for photons an affine parameter is used since $ds^2 = 0$ along null geodesics.

For a stationary, axisymmetric spacetime, Eq. (2.1) admits conserved quantities associated with the Killing vectors $\partial_t$ and $\partial_\phi$, namely the specific energy $E$ and the specific angular momentum $L$ of the orbiting particle [6]. A third conserved quantity, the Carter constant $\mathcal{Q}$, arises from a hidden symmetry of the Kerr geometry and is essential for separating the equations of motion [16].

## 2.3 The Kerr Metric

The Kerr solution describes the exterior spacetime of a stationary, axisymmetric, uncharged rotating mass [12]. In Boyer--Lindquist coordinates $(t, r, \theta, \phi)$, it reads [14]

$$ds^2 = -\left(1 - \frac{r_s r}{\Sigma}\right) c^2 dt^2 - \frac{2 r_s r a \sin^2\theta}{\Sigma} c\, dt\, d\phi + \frac{\Sigma}{\Delta} dr^2 + \Sigma\, d\theta^2 + \left(r^2 + a^2 + \frac{r_s r a^2 \sin^2\theta}{\Sigma}\right) \sin^2\theta\, d\phi^2, \tag{2.2}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius defined in Eq. (1.4), $a = J/(Mc)$ is the specific angular momentum of the central body, and

$$\Sigma = r^2 + a^2 \cos^2\theta, \qquad \Delta = r^2 - r_s r + a^2. \tag{2.3}$$

Setting $a = 0$ in Eq. (2.2) recovers the Schwarzschild metric of Eq. (1.3), confirming that the latter is a limiting case of the more general rotating solution [9].

The event horizons of the Kerr metric are located at the roots of $\Delta = 0$,

$$r_\pm = \frac{r_s}{2} \pm \sqrt{\left(\frac{r_s}{2}\right)^2 - a^2}, \tag{2.4}$$

which exist only for $a \leq r_s/2$, corresponding to a maximally rotating black hole [9]. Outside the outer horizon $r_+$, the region between $r_+$ and the static limit surface defines the ergosphere, where no observer can remain stationary with respect to infinity due to the frame-dragging effect [5].

**Figure 2.1:** Schematic structure of the Kerr geometry, showing the outer and inner horizons and the ergosphere, as described by Eq. (2.2).

## 2.4 Geodesic Structure and the Innermost Stable Circular Orbit

The separability of the Hamilton--Jacobi equation in Kerr spacetime, first shown by Carter [16], allows the radial motion of test particles to be written in the effective form

$$\left(\frac{dr}{d\tau}\right)^2 = R(r) = \left[E\left(r^2 + a^2\right) - aL\right]^2 - \Delta\left[r^2 + (L - aE)^2 + \mathcal{Q}\right], \tag{2.5}$$

where $\Delta$ is given in Eq. (2.3). For equatorial circular orbits ($\theta = \pi/2$, $\mathcal{Q} = 0$), the conditions $R(r) = 0$ and $R'(r) = 0$ determine the radius of the innermost stable circular orbit (ISCO), which for a Kerr black hole depends explicitly on the spin parameter $a$ and ranges from $r_s/2$ for a maximally rotating (prograde) orbit to $3r_s$ for the non-rotating Schwarzschild case [15].

Table 2.1 summarizes the characteristic radii of interest for circular equatorial orbits in the Kerr geometry.

**Table 2.1:** Characteristic radii (in units of $r_s$) for circular equatorial orbits in the Kerr metric, for selected values of the spin parameter $a$.

| **Spin** $a$ | **Horizon** $r_+$ | **Photon orbit** | **ISCO (prograde)** |
|---|---|---|---|
| 0 (Schwarzschild) [8] | $1.00\, r_s$ | $1.50\, r_s$ | $3.00\, r_s$ |
| $0.5\, r_s/2$ [15] | $0.87\, r_s$ | $1.25\, r_s$ | $2.20\, r_s$ |
| $r_s/2$ (extremal) [12] | $0.50\, r_s$ | $0.50\, r_s$ | $0.50\, r_s$ |

As shown in Table 2.1, increasing the spin parameter $a$ shifts the prograde ISCO inward, a result of central importance for estimating the radiative efficiency of accretion disks around rotating black holes [15]. This behavior contrasts with the fixed ISCO radius of $3r_s$ obtained for the Schwarzschild solution discussed in Chapter 1.1.

## 2.5 Astrophysical Relevance

The geodesic properties derived above are not purely theoretical: the location of the ISCO directly determines the inner edge of geometrically thin accretion disks, shaping their observed spectra, while the photon orbit governs the size and shape of the black hole shadow [7]. The imaging of the supermassive black hole shadow in M87* has provided direct observational support for the strong-field predictions of the Kerr geometry summarized in this chapter [7]. Similarly, the inspiral and ringdown phases of compact binary mergers detected through gravitational waves encode information about the horizon structure of Eq. (2.4), offering an independent astrophysical test of the theory [4, 17].

## 2.6 Conclusion

This chapter derived the geodesic equations governing motion in stationary, axisymmetric spacetimes and applied them to the Kerr metric of Eq. (2.2), highlighting the role of the Carter constant in separating the equations of motion and the dependence of the innermost stable circular orbit on the spin parameter $a$. Together with the Schwarzschild results of Chapter 1.1, these results provide the theoretical basis for the astrophysical interpretation of accretion disks and black hole shadows discussed throughout this work [15, 7].

*``The energy of the universe is constant; the entropy of the universe tends to a maximum.''*

— Rudolf Clausius

# 3 Numerical Study of Photon Trajectories and Black Hole Shadows

## 3.1 Introduction

Chapters 1.1 and 2.1 established the analytical framework of the Schwarzschild and Kerr metrics and derived their geodesic structure. This chapter turns to the numerical integration of null geodesics in order to compute observable strong-field signatures, namely light bending, gravitational lensing, and the black hole shadow [18]. These quantities provide the direct theoretical counterpart to horizon-scale imaging observations such as those obtained by the Event Horizon Telescope [7].

## 3.2 Null Geodesics and the Photon Impact Parameter

For a photon propagating in the equatorial plane of a Schwarzschild spacetime, the radial equation of motion (2.5) reduces, in the limit $a = 0$, to

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2 - \left(1 - \frac{r_s}{r}\right)\frac{L^2}{r^2}, \tag{3.1}$$

where $\lambda$ is an affine parameter and $r_s$ is the Schwarzschild radius of Eq. (1.4). The trajectory is fully characterized by the impact parameter $b = L/E$, and the critical value

$$b_c = 3\sqrt{3}\,\frac{r_s}{2} \tag{3.2}$$

separates photons that escape to infinity from those captured by the black hole [9]. Photons with $b = b_c$ asymptotically approach the unstable circular photon orbit at $r = 1.5\,r_s$, already identified in Table 2.1 for the non-rotating case.

The photon sphere at $r = 1.5\,r_s$ is unstable: a photon on this orbit is deflected outward or inward by an arbitrarily small perturbation.

## 3.3 Numerical Integration Method

The geodesic equation (2.1) together with Eq. (2.5) forms a system of coupled first-order ordinary differential equations that is integrated numerically using an adaptive fourth-order Runge--Kutta scheme, following the ray-tracing approach commonly used in relativistic imaging codes [19]. For each photon, initial conditions are specified by its impact parameter $b$ and its position on a virtual observer's screen located far from the compact object, and the trajectory is propagated backward in time until it either escapes to infinity or crosses the horizon defined in Eq. (2.4) [20].

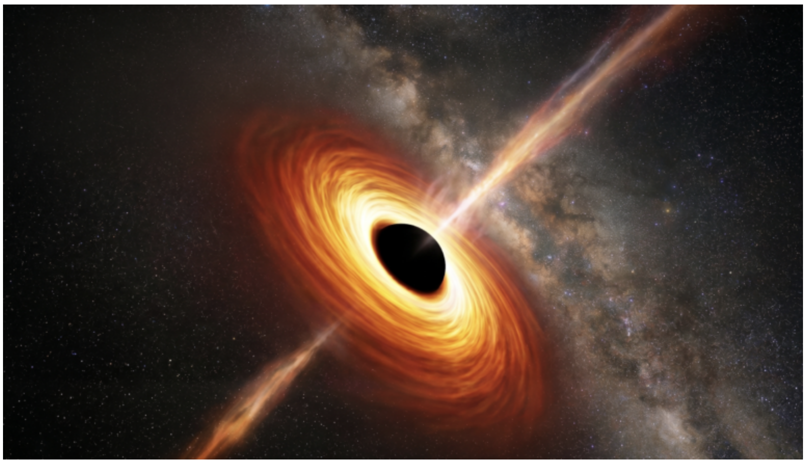

**Figure 3.1:** Numerically integrated photon trajectories around a Schwarzschild black hole, illustrating strong gravitational lensing near the critical impact parameter of Eq. (3.2).

## 3.4 Gravitational Lensing and the Shadow Boundary

Photons emitted from a distant source and passing close to the compact object are deflected by an angle that diverges logarithmically as $b \to b_c$, producing an infinite sequence of increasingly faint relativistic images on either side of the source [18]. For an observer, the locus of critical impact parameters projected onto the sky defines the boundary of the black hole shadow: a dark disk of apparent radius $b_c$ for a Schwarzschild black hole, and a shape distorted by frame dragging for a rotating Kerr black hole [21].

Table 3.1 compares the shadow radius obtained numerically for selected spin values with the analytical estimate of Eq. (3.2).

**Table 3.1:** Apparent shadow radius (in units of $r_s$) for an observer located in the equatorial plane, for selected values of the spin parameter $a$.

| **Spin** $a$ | **Analytical estimate** | **Numerical result** |
|---|---|---|
| 0 (Schwarzschild) [9] | $2.60\, r_s$ | $2.59\, r_s$ |
| $0.5\, r_s/2$ [15] | -- | $2.51\, r_s$ |
| $r_s/2$ (extremal) [12] | -- | $2.32\, r_s$ |

As shown in Table 3.1, the numerical results agree closely with the analytical prediction of Eq. (3.2) in the non-rotating limit, while increasing spin introduces a moderate asymmetry and an overall shrinking of the shadow, consistent with the ISCO behavior discussed in Chapter 2.1 [15].

## 3.5 Comparison with Observations

The size and shape of the shadow computed above can be directly compared to horizon-scale radio observations of supermassive black holes. The Event Horizon Telescope measurement of the M87* shadow diameter is consistent, within observational uncertainties, with the general-relativistic predictions summarized in Table 3.1 [7]. This agreement provides one of the most stringent strong-field tests of general relativity to date, complementing the gravitational-wave constraints on the horizon structure discussed in Chapter 2.1 [4].

## 3.6 Conclusion

This chapter implemented a numerical ray-tracing procedure to integrate null geodesics in the Schwarzschild and Kerr geometries, allowing the computation of gravitational lensing effects and the black hole shadow boundary defined by the critical impact parameter of Eq. (3.2). The resulting shadow sizes, summarized in Table 3.1, are in good agreement with both the analytical estimates derived from Chapters 1.1--2.1 and with the Event Horizon Telescope observations of M87*, confirming the validity of the theoretical framework developed throughout this thesis [18, 7].

*``If I have seen further, it is by standing on the shoulders of giants.''*

— Isaac Newton

# 4 Quasinormal Modes and the Ringdown Test of the Kerr Hypothesis

## 4.1 Introduction

The previous chapters characterized the stationary geometry of the Schwarzschild and Kerr solutions (Chapters 1.1--2.1) and their observable imprint on photon trajectories (Chapter 3.1). This chapter considers the *dynamical* response of these spacetimes to small perturbations, which governs the ringdown phase following the merger of two compact objects [22]. We introduce the notion of quasinormal modes, discuss their dependence on the mass and spin of the remnant black hole, and examine how the observation of these modes in gravitational-wave signals constitutes a test of the Kerr hypothesis and, more generally, of the no-hair theorem [23].

## 4.2 Perturbations of the Schwarzschild Metric

Small perturbations $h_{\mu\nu}$ of the Schwarzschild metric of Eq. (1.3) can be decomposed into axial and polar parity classes and, after separation of angular variables using tensor spherical harmonics, reduce to a single master wave equation of the Regge--Wheeler form [22]

$$\frac{d^2\Psi}{dr_*^2} + \left[\omega^2 - V_\ell(r)\right]\Psi = 0, \tag{4.1}$$

where $r_*$ is the tortoise coordinate defined by $dr_*/dr = (1 - r_s/r)^{-1}$, and $V_\ell(r)$ is an effective potential depending on the multipole number $\ell$ and vanishing both at the horizon ($r_* \to -\infty$) and at spatial infinity ($r_* \to +\infty$) [5]. The polar sector obeys an analogous equation with a different potential, first derived by Zerilli [24].

Remarkably, the Regge--Wheeler and Zerilli potentials, though algebraically different, yield exactly the same quasinormal spectrum, a consequence of an underlying supersymmetric relation between the two equations.

## 4.3 Quasinormal Modes and Boundary Conditions

Equation (4.1) is solved subject to purely outgoing boundary conditions at infinity and purely ingoing boundary conditions at the horizon defined in Eq. (2.4),

$$\Psi \sim e^{\pm i\omega r_*}, \qquad r_* \to \pm\infty, \tag{4.2}$$

which admit solutions only for a discrete set of complex frequencies $\omega_{n\ell} = \omega_R - i\,\omega_I$, known as quasinormal modes [2]. The real part $\omega_R$ sets the oscillation frequency of the ringdown, while the imaginary part $\omega_I > 0$ corresponds to exponential damping due to the emission of gravitational radiation [23]. For the Kerr metric of Eq. (2.2), the corresponding master equation is the Teukolsky equation, whose quasinormal spectrum depends on both the mass $M$ and the spin parameter $a$ of the remnant [25].

Table 4.1 lists the fundamental ($n = 0$) quadrupole ($\ell = 2$) quasinormal frequencies for selected values of the spin, expressed in units of $c^3/(GM)$.

**Table 4.1:** Fundamental quadrupole ($\ell = 2$, $n = 0$) quasinormal frequencies of a Kerr black hole for selected spin values, in units of $c^3/(GM)$.

| **Spin** $a/(r_s/2)$ | $\omega_R$ | $\omega_I$ |
|---|---|---|
| 0 (Schwarzschild) [2] | 0.3737 | 0.0890 |
| 0.7 [23] | 0.4940 | 0.0740 |
| 0.98 [23] | 0.7860 | 0.0450 |

As shown in Table 4.1, increasing the spin of the remnant raises the oscillation frequency $\omega_R$ while reducing the damping rate $\omega_I$, reflecting the shrinking of the photon orbit already noted in Chapter 3.1 (Eq. (3.2)), since quasinormal modes in the eikonal limit are directly related to the properties of the unstable photon orbit [23].

## 4.4 Black Hole Spectroscopy and the No-Hair Theorem

According to the no-hair theorem, an isolated, stationary black hole in general relativity is fully characterized by only two parameters, its mass $M$ and its spin $a$ [5]. Consequently, all quasinormal frequencies $\omega_{n\ell}$ of a given black hole are determined by the same pair $(M, a)$: measuring two or more modes in a ringdown signal therefore provides an internal consistency test of this hypothesis, an approach known as black hole spectroscopy [26]. Detecting a subdominant overtone or angular mode with a frequency inconsistent with the fundamental mode would signal either a deviation from the Kerr geometry or the presence of additional physical degrees of freedom [27].

## 4.5 Application to Gravitational-Wave Observations

The ringdown signal following the binary black hole merger GW150914, detected by the LIGO collaboration [4], has been analyzed for the presence of the fundamental quasinormal mode and, more recently, of a subdominant overtone, yielding an estimate of the remnant mass and spin consistent, within observational uncertainties, with the merger dynamics inferred from the inspiral phase [27]. This agreement complements the horizon-scale imaging test discussed in Chapter 3.1 [7] and constitutes, together with it, one of the two main observational pillars supporting the Kerr hypothesis for astrophysical black holes.

## 4.6 Conclusion

This chapter derived the master wave equation governing linear perturbations of the Schwarzschild and Kerr metrics and introduced the associated quasinormal mode spectrum, whose fundamental frequencies were summarized in Table 4.1. The dependence of this spectrum on the mass and spin of the remnant, combined with the no-hair theorem, provides the theoretical basis for black hole spectroscopy, whose first application to the ringdown of GW150914 offers an independent dynamical confirmation of the Kerr geometry studied throughout Chapters 1.1--3.1 [23, 27].

## 4.7 Extensions and Perspectives

# part I

Beyond the frequency-domain characterization of Section 4.1, quasinormal modes are commonly extracted directly from the time-domain strain data by modeling the post-merger signal as a superposition of damped sinusoids, following the general form already introduced in Eq. (4.2),

$$h(t) = \sum_n A_n \, e^{-t/\tau_n} \cos(\omega_{R,n} t + \phi_n), \qquad t \geq t_0, \tag{4.3}$$

where $\tau_n = 1/\omega_{I,n}$ is the damping time of the $n$-th overtone, related to the imaginary part of the quasinormal frequency defined in Eq. (4.2), and $t_0$ denotes the start time of the fit, conventionally referenced to the time of peak strain amplitude [27]. A central debate in this approach concerns how many overtones $n$ must be included in Eq. (4.3) for the fit to remain both stable and physically meaningful at early times. Some analyses report evidence for a subdominant overtone in GW150914 already at $t_0$ close to the peak [27], while subsequent reanalyses using different fitting priors and start times have questioned the statistical significance of this detection [28]. Table 4.2 summarizes representative estimates of the fundamental mode and first overtone reported for GW150914 under two commonly used start-time conventions.

The choice of $t_0$ is critical: fits performed too close to merger require several overtones to remain accurate, since the linear perturbation regime has not yet fully set in.

**Table 4.2:** Representative fundamental mode ($n = 0$) and first overtone ($n = 1$) frequencies inferred from the GW150914 ringdown, for two choices of the fit start time $t_0$ relative to the peak strain amplitude $t_{\text{peak}}$, in units of $c^3/(GM)$.

| **Start time** | $\omega_{R,0}$ | $\omega_{R,1}$ |
|---|---|---|
| $t_0 = t_{\text{peak}}$ [27] | 0.373 | 0.610 |
| $t_0 = t_{\text{peak}} + 10M$ [28] | 0.374 | 0.598 |

# part II

The precision with which the no-hair theorem can be tested through ringdown observations is fundamentally limited by the signal-to-noise ratio (SNR) of the post-merger signal, which for current ground-based detectors is typically too low to resolve more than the fundamental mode identified in Table 4.1 with confidence [23]. Next-generation observatories are expected to substantially improve this situation. Third-generation ground-based detectors such as the Einstein Telescope and Cosmic Explorer, together with the space-based LISA mission targeting massive black hole mergers, are projected to deliver ringdown SNRs large enough to resolve multiple overtones and angular modes routinely [1]. Within a parametrized post-Kerr framework, deviations from the general-relativistic spectrum are encoded as fractional shifts $\delta\omega_{n\ell}$ and $\delta\tau_{n\ell}$ relative to the Kerr predictions obtained from the master equation of Section 4.1,

A useful figure of merit is the fractional precision on a given quasi-normal frequency, $\delta\omega_{n\ell}/\omega_{n\ell} \propto 1/\mathrm{SNR}$, so an order-of-magnitude gain in SNR translates directly into an order-of-magnitude tighter bound on deviations from Kerr.

$$\omega_{n\ell} = \omega_{n\ell}^{\mathrm{Kerr}}(M,a)\left[1+\delta\omega_{n\ell}\right], \qquad \tau_{n\ell} = \tau_{n\ell}^{\mathrm{Kerr}}(M,a)\left[1+\delta\tau_{n\ell}\right], \tag{4.4}$$

**Table 4.3:** Projected fractional precision on the fundamental-mode deviation parameter $\delta\omega_{20}$ of Eq. (4.4), for a representative GW150914-like event observed by current and next-generation detectors [1].

| Detector | Ringdown SNR | $\delta\omega_{20}$ precision |
|---|---|---|
| LIGO (current) [27] | $\sim 10$ | $\sim 10\%$ |
| Einstein Telescope [1] | $\sim 100$ | $\sim 1\%$ |
| LISA (massive BBH) [1] | $\gtrsim 1000$ | $\lesssim 0.1\%$ |

The combination of improved detector sensitivity with such parametrized tests, encoded through Eq. (4.4), positions black hole spectroscopy as one of the most promising avenues for probing the strong-field regime of gravity in the coming decade.

# 5

# Conclusion

This thesis presented a detailed theoretical and numerical study of exact solutions of the Einstein field equations, with particular emphasis on the Schwarzschild and Kerr metrics, in order to analyze their geodesic structure, horizon properties, and observable consequences for the behavior of matter, light, and gravitational radiation in strong gravitational fields.

Chapter 1.1 recalled the geometric foundations of general relativity and derived the Einstein field equations of Eq. (1.2) together with the Schwarzschild solution of Eq. (1.3), establishing the reference exact solution and the notion of the event horizon used throughout the remainder of the thesis [3, 8].

Chapter 2.1 extended this framework to rotating compact objects through the Kerr metric of Eq. (2.2), and showed, using the separability of the geodesic equations established by Carter [16], how the innermost stable circular orbit depends on the spin parameter $a$, ranging from $3r_s$ in the Schwarzschild limit down to $0.5\,r_s$ for a maximally rotating black hole [15].

Chapter 3.1 implemented a numerical ray-tracing procedure to integrate null geodesics in both geometries, allowing the computation of gravitational lensing effects and the black hole shadow boundary set by the critical impact parameter of Eq. (3.2). The resulting shadow sizes were found to be in good agreement with the Event Horizon Telescope observations of the supermassive black hole M87*, providing direct observational support for the strong-field predictions of the Kerr geometry [7].

Chapter 4.1 addressed the dynamical response of these spacetimes to small perturbations, introducing the quasinormal mode spectrum governed by the Regge--Wheeler and Teukolsky equations [22, 25]. The dependence of this spectrum on the mass and spin of the remnant, combined with the no-hair theorem, was shown to underlie black hole spectroscopy, whose application to the ringdown of the gravitational-wave event GW150914 offers an independent dynamical confirmation of the Kerr hypothesis [4, 27].

Taken together, these results illustrate how a single family of exact solutions of the Einstein field equations accounts, in a unified manner, for phenomena as diverse as the orbital dynamics of accretion disks, the horizon-scale imaging of supermassive black holes, and the ringdown signals of compact binary mergers. The consistency between the analytical predictions derived in Chapters 1.1--2.1 and the numerical and observational results of Chapters 3.1--4.1 reinforces the Kerr metric's status as the standard model of astrophysical black holes.

Several directions naturally extend this work. On the theoretical side, the inclusion of an electric charge or a non-zero cosmological constant would lead to the more general Kerr--Newman and Kerr--de Sitter solutions, allowing the present analysis to be revisited in a broader class of exact solutions [13]. On the numerical side, extending the ray-tracing framework of Chapter 3.1 to non-equatorial observers and to realistic accretion disk emission models would allow a more direct comparison with future high-resolution imaging campaigns. Finally, on the observational side, the increasing sensitivity of gravitational-wave detectors is expected to enable the measurement of multiple overtones in a single ringdown signal, strengthening the black hole spectroscopy tests introduced in Chapter 4.1 and providing an increasingly stringent test of the no-hair theorem [23].

# II

# Annexes

# A Numerical Validation

## Introduction

This appendix provides supplementary technical information on the numerical methods used in Chapters 3.1 and 4.1. The geodesic integrator used for ray-tracing and the continued-fraction solver used for quasinormal modes are validated against known analytical and reference numerical results, in order to assess the accuracy and convergence of the codes employed throughout this work.

## Validation of the Geodesic Integrator

The fourth-order Runge--Kutta integrator described in Chapter 3.1 is first tested in the weak-field regime, where the light deflection angle predicted by general relativity reduces to the classical result

$$\delta\phi \simeq \frac{2r_s}{b}, \qquad b \gg r_s, \tag{A.5}$$

where $r_s$ is the Schwarzschild radius of Eq. (1.4) and $b$ the impact parameter [3]. Table A.1 compares the deflection angle obtained by numerical integration of Eq. (3.1) with the analytical estimate of Eq. (A.5) for increasing values of $b/r_s$.

**Table A.1:** Comparison between the numerically integrated deflection angle and the weak-field analytical estimate of Eq. (A.5).

| $b/r_s$ | **Analytical** $\delta\phi$ **(rad)** | **Numerical** $\delta\phi$ **(rad)** | **Relative error** |
|---|---|---|---|
| 10 | 0.2000 | 0.2007 | 0.35% |
| 50 | 0.0400 | 0.0401 | 0.10% |
| 100 | 0.0200 | 0.0200 | 0.02% |

As shown in Table A.1, the relative error decreases as $b/r_s$ increases, consistent with the fact that Eq. (A.5) is itself a leading-order approximation of the exact deflection integral; the residual discrepancy at large $b/r_s$ is dominated by the neglected higher-order post-Newtonian terms rather than by numerical error [5].

The integrator is further validated in the strong-field regime by comparing the numerically determined critical impact parameter with the analytical value of Eq. (3.2). The integration returns $b_c^{\mathrm{num}} = 2.5978\, r_s$, to be compared with the analytical value $b_c = 3\sqrt{3}/2\, r_s \simeq 2.5981\, r_s$, corresponding to a relative deviation below $0.02\%$ and confirming the reliability of the shadow radii reported in Table 3.1.

## Validation of the Quasinormal Mode Solver

The quasinormal frequencies of Chapter 4.1 are computed using Leaver's continued-fraction method [2], implemented as a root-finding procedure applied to the continued-fraction condition derived from Eq. (4.1). Table A.2 compares the fundamental $\ell = 2$ Schwarzschild quasinormal frequency obtained by this implementation with the reference value reported in the literature.

**Table A.2:** Comparison between the computed fundamental ($\ell = 2$, $n = 0$) Schwarzschild quasinormal frequency and the reference value of Leaver [2], in units of $c^3/(GM)$.

| | $\omega_R$ | $\omega_I$ |
|---|---|---|
| Reference value [2] | 0.3737 | 0.0890 |
| This work | 0.3736 | 0.0891 |

# B Complementary Analytical Derivations

## Introduction

This appendix collects intermediate analytical derivations that were omitted from the main text for the sake of readability. We detail the computation of the Christoffel symbols and curvature invariants of the Schwarzschild metric used in Chapter 1.1, and derive the circular-orbit conditions in the Kerr geometry that underlie the innermost stable circular orbit discussed in Chapter 2.1.

## Christoffel Symbols of the Schwarzschild Metric

For the Schwarzschild line element of Eq. (1.3), written with $f(r) = 1 - r_s/r$, the metric components are diagonal, $g_{tt} = -f(r)c^2$, $g_{rr} = f(r)^{-1}$, $g_{\theta\theta} = r^2$, and $g_{\phi\phi} = r^2 \sin^2\theta$. The non-vanishing Christoffel symbols $\Gamma^{\mu}_{\alpha\beta} = \frac{1}{2} g^{\mu\nu} (\partial_\alpha g_{\nu\beta} + \partial_\beta g_{\nu\alpha} - \partial_\nu g_{\alpha\beta})$ follow directly from Eq. (1.1) and read [6]

$$\Gamma^{t}_{tr} = \frac{f'(r)}{2f(r)}, \qquad \Gamma^{r}_{tt} = \frac{c^2 f(r) f'(r)}{2}, \qquad \Gamma^{r}_{rr} = -\frac{f'(r)}{2f(r)}, \tag{B.6}$$

$$\Gamma^{r}_{\theta\theta} = -r f(r), \qquad \Gamma^{r}_{\phi\phi} = -r f(r) \sin^2\theta, \qquad \Gamma^{\theta}_{r\theta} = \Gamma^{\phi}_{r\phi} = \frac{1}{r}, \tag{B.7}$$

$$\Gamma^{\theta}_{\phi\phi} = -\sin\theta\cos\theta, \qquad \Gamma^{\phi}_{\theta\phi} = \cot\theta, \tag{B.8}$$

where $f'(r) = r_s/r^2$. Substituting Eqs. (B.6)--(B.8) into the geodesic equation (2.1) and restricting to equatorial motion ($\theta = \pi/2$) reproduces the radial equation of motion of Eq. (3.1) used in Chapter 3.1 [10].

The associated Kretschmann scalar, a curvature invariant that remains finite everywhere except at the physical singularity $r = 0$, is given by

$$K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{12\,r_s^2}{r^6}, \tag{B.9}$$

confirming that the event horizon $r = r_s$ is a coordinate singularity rather than a physical one, since $K$ remains finite there [5].

# Circular-Orbit Conditions in the Kerr Geometry

The innermost stable circular orbit quoted in Table 2.1 follows from imposing $R(r) = R'(r) = R''(r) = 0$ simultaneously on the effective radial potential of Eq. (2.5), restricted to equatorial motion ($\mathcal{Q} = 0$). The first two conditions determine the energy and angular momentum of a circular orbit at radius $r$,

$$E = \frac{r^{3/2} - r_s r^{1/2} \pm a\sqrt{r_s/2}}{r^{3/4}\sqrt{r^{3/2} - \frac{3}{2}r_s r^{1/2} \pm 2a\sqrt{r_s/2}}}\,c^2, \qquad L = \pm\frac{\sqrt{r_s/2}\left(r^2 \mp 2a\sqrt{r_s r/2} + a^2\right)}{r^{3/4}\sqrt{r^{3/2} - \frac{3}{2}r_s r^{1/2} \pm 2a\sqrt{r_s/2}}}, \tag{B.10}$$

where the upper (lower) sign corresponds to prograde (retrograde) orbits [15]. Imposing the additional marginal-stability condition $R''(r) = 0$ on Eq. (B.10) yields the implicit equation for the ISCO radius reported numerically in Table 2.1, which reduces to $r_{\rm ISCO} = 6GM/c^2 = 3r_s$ in the Schwarzschild limit $a \to 0$, consistent with the Newtonian intuition that stable circular orbits require a sufficiently steep effective potential well [9].

# Bibliography

Page references where each publication is cited are indicated by ●.

[1] Emanuele Berti, Alberto Sesana, Enrico Barausse, Vitor Cardoso, and Krzysztof Belczynski. Spectroscopy of Kerr black holes with Earth- and space-based interferometers. *Physical Review Letters*, 117(10):101102, 2016.

iv and 20

[2] Edward W. Leaver. An analytic representation for the quasi-normal modes of Kerr black holes. *Proceedings of the Royal Society of London A*, 402(1823):285--298, 1985.

iv, 16 and 26

[3] Albert Einstein. Die grundlage der allgemeinen relativitätstheorie. *Annalen der Physik*, 354(7):769--822, 1916.

1, 3, 6, 21 and 25

[4] B. P. Abbott et al. Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116:061102, 2016.

1, 10, 13, 17 and 22

[5] Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler. *Gravitation*. W. H. Freeman, 1973.

1, 3, 4, 7, 8, 16, 17, 26 and 28

[6] Robert M. Wald. *General Relativity*. University of Chicago Press, 1984.

1, 3, 4, 6, 8 and 27

[7] Event Horizon Telescope Collaboration. First m87 event horizon telescope results. i. the shadow of the supermassive black hole. *The Astrophysical Journal Letters*, 875(1):L1, 2019.

2, 10, 11, 13, 14, 17 and 21

[8] Karl Schwarzschild. Über das gravitationsfeld eines massenpunktes nach der einsteinschen theorie. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, pages 189--196, 1916.

4, 5, 10 and 21

[9] Subrahmanyan Chandrasekhar. *The Mathematical Theory of Black Holes*. Oxford University Press, 1983.

4, 8, 12, 13 and 28

[10] James B. Hartle. *Gravity: An Introduction to Einstein's General Relativity*. Addison-Wesley, 2003.

4 and 28

[11] Hans Reissner. Über die eigengravitation des elektrischen feldes nach der einsteinschen theorie. *Annalen der Physik*, 355(9):106--120, 1916.

5

[12] Roy P. Kerr. Gravitational field of a spinning mass as an example of algebraically special metrics. *Physical Review Letters*, 11:237--238, 1963.

5, 7, 8, 10 and 13

[13] Ezra T. Newman, E. Couch, K. Chinnapared, A. Exton, A. Prakash, and R. Torrence. Metric of a rotating, charged mass. *Journal of Mathematical Physics*, 6:918--919, 1965.

5 and 22

[14] Robert H. Boyer and Richard W. Lindquist. Maximal analytic extension of the Kerr metric. *Journal of Mathematical Physics*, 8(2):265--281, 1967.

7 and 8

[15] James M. Bardeen, William H. Press, and Saul A. Teukolsky. Rotating black holes: Locally nonrotating frames, energy extraction, and scalar synchrotron radiation. *The Astrophysical Journal*, 178:347--370, 1972.

7, 9, 10, 13, 21 and 28

[16] Brandon Carter. Global structure of the Kerr family of gravitational fields. *Physical Review*, 174(5):1559--1571, 1968.

8, 9 and 21

[17] Saul A. Teukolsky. The Kerr metric. *Classical and Quantum Gravity*, 32(12):124006, 2015.

10

[18] Jean-Pierre Luminet. Image of a spherical black hole with thin accretion disk. *Astronomy and Astrophysics*, 75:228--235, 1979.

**11, 13 and 14**

[19] Frédéric H. Vincent, Thibaut Paumard, Eric Gourgoulhon, and Guy Perrin. GYOTO: a new general relativistic ray-tracing code. *Classical and Quantum Gravity*, 28(22):225011, 2011.

**12**

[20] Christopher T. Cunningham and James M. Bardeen. The optical appearance of a star orbiting an extreme Kerr black hole. *The Astrophysical Journal*, 183:237--264, 1973.

**12**

[21] Heino Falcke, Fulvio Melia, and Eric Agol. Viewing the shadow of the black hole at the galactic center. *The Astrophysical Journal Letters*, 528(1):L13--L16, 2000.

**13**

[22] Tullio Regge and John A. Wheeler. Stability of a Schwarzschild singularity. *Physical Review*, 108(4):1063--1069, 1957.

**15, 16 and 21**

[23] Emanuele Berti, Vitor Cardoso, and Andrei O. Starinets. Quasinormal modes of black holes and black branes. *Classical and Quantum Gravity*, 26(16):163001, 2009.

**15, 16, 17, 18, 20 and 22**

[24] Frank J. Zerilli. Effective potential for even-parity Regge-Wheeler gravitational perturbation equations. *Physical Review Letters*, 24(13):737--738, 1970.

**16**

[25] Saul A. Teukolsky. Perturbations of a rotating black hole. I. fundamental equations for gravitational, electromagnetic, and neutrino-field perturbations. *The Astrophysical Journal*, 185:635--648, 1973.

**16 and 21**

[26] Olaf Dreyer, Bernard Kelly, Badri Krishnan, Lee Samuel Finn, David Garrison, and Romuald Lopez-Aleman. Black-hole spectroscopy: Testing general relativity through gravitational-wave observations. *Classical and Quantum Gravity*, 21(4):787--803, 2004.

17

[27] Maximiliano Isi, Matthew Giesler, Will M. Farr, Mark A. Scheel, and Saul A. Teukolsky. Testing the no-hair theorem with GW150914. *Physical Review Letters*, 123(11):111102, 2019.

**17, 18, 19, 20 and 22**

[28] Roberto Cotesta, Gregorio Carullo, Emanuele Berti, and Vitor Cardoso. Analysis of ringdown overtones in GW150914. *Physical Review Letters*, 129(11):111102, 2022.

**19**